DÉMO · CHANGEMENTS ENTRE LES ÉTAPES

# Cycle de vie et optimisations

Chaque étape comparée à la précédente : ce qui est ajouté et ce qui est retiré.

`+ ligne ajoutée` `− ligne retirée` `@@ position dans le fichier`

1. **Étape 1 → 2** · etape-2-react-memo (2 fichiers)
2. **Étape 2 → 3** · etape-3-usecallback-usememo (2 fichiers)

## Étape 1 → 2 etape-1-cycle-logs → etape-2-react-memo

React.memo et le state seeMore pour observer les renders inutiles

COMPARÉ **src/components/Cars.tsx → src/components/CarsMemo.tsx** +12 −2

```
@@ -1,3 +1,4 @@
+ import React from "react";
  import { useEffect, useRef, useState } from "react";

  function Car({ make, color }) {
@@ -24,6 +25,8 @@
    );
  }

+ const MemoizedCar = React.memo(Car);
+
  const allCars = [
    { make: "BMW", color: "Blue" },
    { make: "Audi", color: "White" },
@@ -36,6 +39,7 @@

    const [cars, setCars] = useState<any>([]);
    const [loading, setLoading] = useState<boolean>(true);
+   const [seeMore, setSeeMore] = useState<boolean>(false);

    console.log(`Home Rendered ${renderCount.current} times`, {
      loading,
@@ -71,7 +75,7 @@

    return (
      <div>
-       <h1>Welcome to the Home Page</h1>
+       <h1>{"Welcome to the Home Page"}</h1>
        <h2>Car List</h2>

        {loading ? (
@@ -80,10 +84,16 @@
          <>
            <div className="container">
              {cars.map((car: any, index: number) => (
-               <Car key={index} make={car.make} color={car.color} />
+               <MemoizedCar key={index} make={car.make} color={car.color} />
              ))}
            </div>
            <button onClick={() => handleFilter("BMW")}>Set Make to BMW</button>
+           <button
+             style={{ margin: "10px" }}
```

```
                onClick={() => setSeeMore(!seeMore)}
              >
                See More
              </button>
            </>
          )}
        </div>
```

MODIFIÉ **src/App.jsx** +1 −1

```
@@ -13,7 +13,7 @@
        <button onClick={() => setShowHome((v) => !v)}>
          {showHome ? "Unmount Home" : "Mount Home"}
        </button>
-       {showHome ? <Home /> : <div>Home is hidden</div>}
+       {showHome ? <CarsMemo /> : <div>Home is hidden</div>}
      </div>
    );
  }
```

## Étape 2 → 3 etape-2-react-memo → etape-3-usecallback-usememo

useCallback (onSelect) et useMemo (carStyle) pour stabiliser les props

COMPARÉ **src/components/CarsMemo.tsx → src/components/CarsUseMemo.tsx** +38 −12

```
@@ -1,31 +1,35 @@
- import React from "react";
+ import React, { useCallback, useMemo } from "react";
  import { useEffect, useRef, useState } from "react";

- function Car({ make, color }) {
+ function Car({ make, color, onSelect, style }: any) {
    const renderCount = useRef(0);
    renderCount.current += 1;
-   let colorInfo = color ? color : "Unknown";

-   console.log(`Car ${make} Rendered ${renderCount.current} times`, {
-     make,
-   });
+   console.log(`Car ${make} Rendered ${renderCount.current} times`);

    useEffect(() => {
      console.log(`Car ${make} Mounted`);
-     return () => {
-       console.log(`Car ${make} UnMounted`);
-     };
+     return () => console.log(`Car ${make} UnMounted`);
    }, [make]);

    return (
-     <div className="car-item">
+     <div className="car-item" style={style} onClick={() => onSelect(make)}>
        <div>{make}</div>
-       <div>{colorInfo}</div>
+       <div>{color ?? "Unknown"}</div>
      </div>
    );
  }

  const MemoizedCar = React.memo(Car);
+
+ /*
+ const MemoizedCar = React.memo(Car, (prevProps, nextProps) => {
+   return (
+     prevProps.brand === nextProps.brand &&
+     prevProps.color === nextProps.color
+   );
+ });
+ */

  const allCars = [
    { make: "BMW", color: "Blue" },
@@ -45,6 +49,22 @@
      loading,
      cars: cars.length,
    });
+
+   // useCallback: même référence entre les renders
+   const onSelect = useCallback((make: string) => {
+     console.log("Selected:", make);
+   }, []);
+
```

```
                58 +   // useMemo: même objet entre les renders
                59 +   const carStyle = useMemo(
                60 +     () => ({
                61 +       padding: "10px",
                62 +       marginBottom: "10px",
                63 +       border: "1px solid #ccc",
                64 +       cursor: "pointer",
                65 +     }),
                66 +     []
                67 +   );
48              68
49              69      useEffect(() => {
50              70        console.log("Home Mounted");
                    @@ -84,7 +104,13 @@
84              104           <>
85              105             <div className="container">
86              106               {cars.map((car: any, index: number) => (
87                  -               <MemoizedCar key={index} make={car.make} color={car.color} />
                107 +               <MemoizedCar
                108 +                 key={index}
                109 +                 make={car.make}
                110 +                 color={car.color}
                111 +                 onSelect={onSelect}
                112 +                 style={carStyle}
                113 +               />
88              114               ))}
89              115             </div>
90              116             <button onClick={() => handleFilter("BMW")}>Set Make to BMW</button>
```

MODIFIÉ **src/App.jsx** +1 −1

```
                    @@ -13,7 +13,7 @@
13              13          <button onClick={() => setShowHome((v) => !v)}>
14              14            {showHome ? "Unmount Home" : "Mount Home"}
15              15          </button>
16                  -       {showHome ? <CarsMemo /> : <div>Home is hidden</div>}
                16 +       {showHome ? <CarsUseMemo /> : <div>Home is hidden</div>}
17              17        </div>
18              18      );
19              19    }
```